

PHASE PLANE TUNING OF FUZZY CONTROLLER FOR 1 DOF HELICOPTER MODEL

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Received 18 December 2014; accepted 20 April 2015

Abstract: The paper suggests a design procedure for fuzzy cascade control. The expert knowledge based tuning procedure is supported by investigation the phase plane of the reference tracking. The plant is a one degree of freedom helicopter model with highly non-linear dynamics in a hardware-in-the-loop arrangement. Starting from a common fuzzy-PI controller in the outer loop and classic PI-controller in the inner loop, the presented tuning procedure enables to obtain fuzzy-adaptive parameter settings of the inner-loop and a heuristic modified rule base in the outer loop. The developed control strategy proves to be very successful: both the settling times and overshoots have become well regulated.

Keywords: Fuzzy control, Phase plane, Pitch control, Cascade control

1. Introduction

Autonomous flight systems require intelligent control strategy. The flight control is a hierarchical system: in the so-called ‘field-level’ usually some simple, separated, classical PID-type control algorithms work. These low level algorithms act directly on the motor drivers. In a holonic point of view these components of the systems can be thought as simple reflective agents that are coordinated and cooperated on higher control levels. The next level above the field level is the set-point control level. In the set-point control level soft control algorithms already can take place. These soft control algorithms are in most cases nested fuzzy controllers. Tuning of these fuzzy controllers is a great challenge for several reasons: the controller has many variable parameters and there are no common methods for optimization or tuning [1], [2], [3], [4].

In this paper the tuning procedure of a fuzzy set-point controller is presented. Initially the controller was a common fuzzy-PI controller that adjusts the set-point of a motor voltage-current control loop of real 1 Degree Of Freedom (DOF) helicopter model as a plant. Instead of applying only software plant simulator, this hardware-in-the-loop arrangement provides more realistic test system. The model is not symmetric and it is non-linear in character and it can be treated as a black-box system. The two nested control loops are in cascade arrangement; where in the inner loop is a classic PI-controller. This loop represents the field-level control. The design of the control system is an iterative procedure based on expert knowledge. This procedure results an adaptive fuzzy decision system (fuzzy-DS) for tuning the proportional gain of the PI-controller in the inner loop, and a non-symmetric modified fuzzy PI-controller in the outer-loop.

The expert knowledge based tuning procedure is supported by investigation the phase plane stability of the fuzzy set-point control. The developed control strategy prove to be much more successful compared to the classic cascade control: both the settling times and overshoots become well controlled. The next section presents phase plane trajectories of a fuzzy PI-controller in a disturbance-free model conditions.

2. Phase plane of a fuzzy PI-controller

The fuzzy PI-controller seems to be a most popular fuzzy controller, it is the first stage moving from the classic PID-controller to a fuzzy controller, and in most cases this can be tuned so, that the control requirement could be satisfied with it [2]. The phase plane method is a graphical method from which information about transient behavior and stability can be obtained by constructing phase plane trajectories [5]. The development of a fuzzy controller is based on expert knowledge, thus there are no exact optimization procedures, only semi-optimal control systems can be achieved: the fuzzy controller is tuned iteratively, until the evolution of the error satisfies the prescribed form and with constant reference signal the process variable remains inside a narrow band around the reference. In this paper the phase plane investigation will be applied for parameter tuning, and for satisfying the quality requirements of the developed fuzzy controller. Usually the phase planes are applied for stability analyses [1], [5] or for initial settings of fuzzy controller [6].

The fuzzy controller has two inputs: error (e) and its discrete derivative (Δe), and one output: the increment of control signal (Δu). The rule base shows symmetry (with different sign) and can be illustrated in table form. For example in *Table I* a typical rule base of a fuzzy PI-controller can be seen, where both inputs have 5 terms: negative big: NB; negative: N; zero: Z; positive: P; positive big: PB; the consequence terms are NEGB; NEG; ZERO; POZ and POZB.

When the system is in stable, error free steady-state, only the rule in the middle of the table is active: IF $e = ZE$ AND $\Delta e = ZE$ THEN $\Delta u = ZERO$. This means that the fix point of a stable control trajectory is at the origin. In this paper first different system trajectories are presented in model conditions for references providing. The plant is assumed as a linear, first order time delayed system. The controlled is a fuzzy PI-controller and the set-point signal (or reference) is generated in a periodic square waveform. The variables are normalized.

Table I

Rule base of a common fuzzy PI-controller

Δe	NB	N	ZE	P	PB
NB	NEGB	NEGB	NEG	NEG	ZERO
N	NEGB	NEG	NEG	ZERO	POZ
ZE	NEG	NEG	ZERO	POZ	POZ
P	NEG	ZERO	POZ	POZ	POZB
PB	ZERO	POZ	POZ	POZB	POZB

Depending on the controller parameterization, different trajectories can be achieved. *Fig. 1* shows a not well damped case in time domain: the process variable follow the reference signal with high overshoot and the transient could not decay in the available time period. The corresponding trajectories show circulations around the fix point lying in the origin. The next two examples in *Fig. 2* and in *Fig. 3* show transients and trajectories for reference follow without overshoot, but remain periodic in control variable around the steady-state.

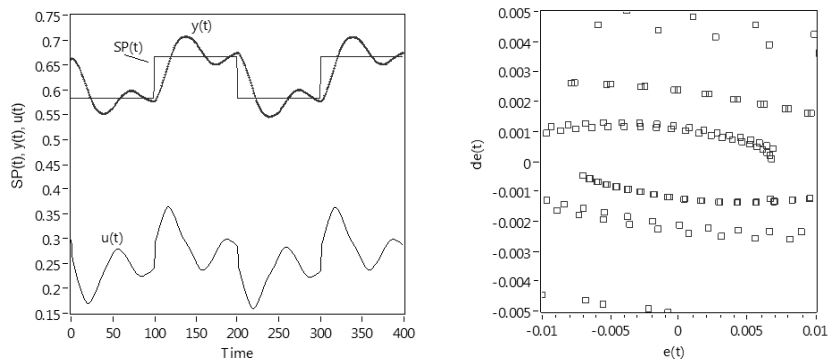


Fig. 1. Set-point tracking with fuzzy Pi-controller, 1. example: no steady-state in the time available. Left: time functions of set-point $SP(t)$, process variable ($y(t)$) and control variable ($u(t)$). Right: corresponding phase-plane trajectories

This can be seen from the step response functions as well, but from time functions the reason of the periodic behavior could not be deduce. In the corresponding trajectories it can be seen that in the first case the error input terms and the second case the error derivative are badly defined around the origin, these membership functions must be corrected (redefine partitioning around the zero). Therefore not only the cause of the inadequate control but the suggestion for improvement can be deducted from the trajectories. *Fig. 4* shows reference following of an appropriately parameterized fuzzy control loop. These trajectories go to the fix point.

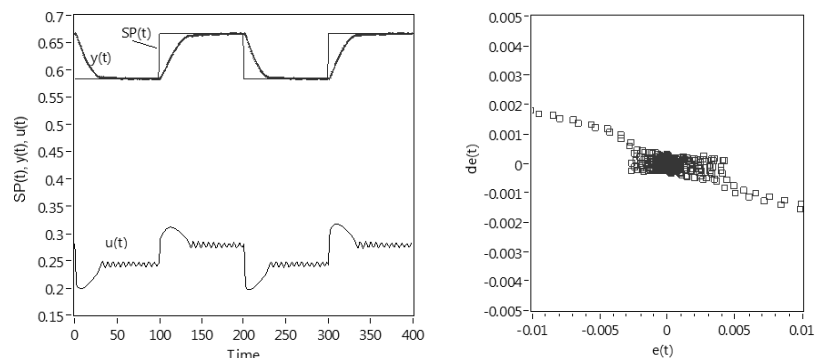


Fig. 2. Set-point tracking with fuzzy Pi-controller, 2. example: there is a periodic change around the set-point in quasi-steady-state. Left: time functions of set-point $SP(t)$, process variable $y(t)$ and control variable $u(t)$. Right: corresponding phase-plane trajectories

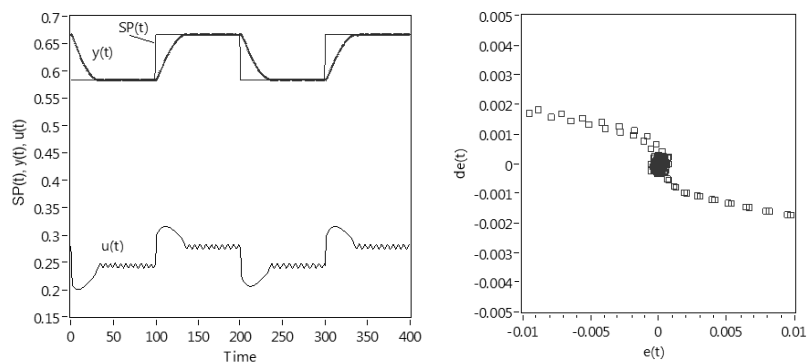


Fig. 3. Set-point tracking with fuzzy Pi-controller, 3. example: there is a periodic change around the set-point in quasi-steady-state. Left: time functions of set-point $SP(t)$, process variable $y(t)$ and control variable $u(t)$. Right: corresponding phase-plane trajectories

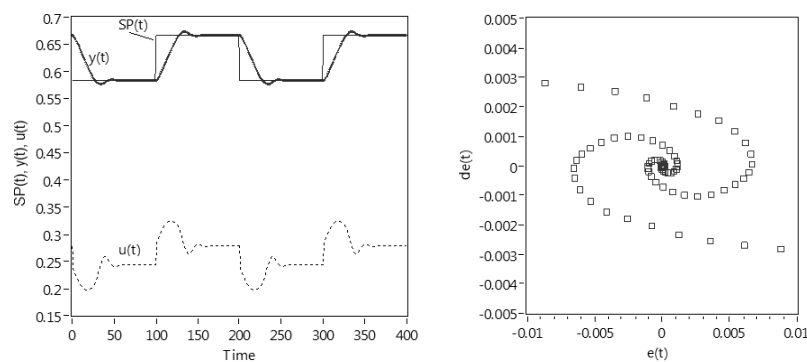


Fig. 4. Set-point tracking with fuzzy Pi-controller, 4. example: appropriate set-point tracking with acceptable overshoot. Left: time functions of set-point $SP(t)$, process variable $y(t)$ and control variable $u(t)$. Right: corresponding phase-plane trajectories

3. Development of a 1-DoF flight control strategy

3.1. Experimental setup

The Quanser Vertical Take-Off and Landing (VTOL) Trainer is a balance like, 1 DoF helicopter model that is designed exclusively for the National Instruments Educational Laboratory Virtual Instrumentation Suite (NI ELVIS). The QNET model is a hardware-in-the-loop implementation: an effective tool for teaching controls fundamentals in various engineering disciplines [7]. The QNET VTOL module introduces students to the fundamentals of aerospace engineering, like basic flight dynamics and control. The model system consists of one variable-speed fan with safety guards mounted on the end of a cantilevered arm. At the other end of the arm an adjustable counterweight is attached. The VTOL device pivots about the pitch axis. The position of the counterweight affects the dynamics of the system [8]. *Fig. 5* shows the experimental setup. Measured process variables are the position (pitch), which is deduced from the signal of a digital encoder mounted at the axis and the armature current of the fan. The Manipulated Variable (MV) is the motor voltage, the power amplifier Pulse-Width Modulated (PWM) is mounted in the test panel. The control algorithm was developed in LabVIEW graphical programming system.



Fig. 5. Quanser VTOL Trainer equipped in NI ELVIS II. DAQ-board. The DAQ-board is connected to a PC with an USB connection

The original control software delivered by the manufacturer is an example for a cascade control: the VTOL device is broken down into two subsystems: the voltage-current dynamics of the motor and the current-position dynamics of the VTOL body. A digital PI controller in the inner loop is designed to regulate the current inside the motor, according to a desired current set-point. This current set-point is generated from the outer-loop controller: a digital PID algorithm controls the pitch of the VTOL trainer. This cascade control with manufacturer settings has been tested in the same control conditions that will be presented further in this paper but this configuration could not satisfy the strict requirement for the reference tracking. This fact led to the decision to apply fuzzy logic offered opportunities in the master loop.

3.2. Investigation of the cascade control loop

The cascade control consists of two nested control loops. In the inner control loop there is a slave controller tuned for the voltage-current dynamics of the motor and in the outer control loop there is a master controller tuned for the current-position dynamics of the VTOL body. This arrangement models the situation of a manual expert control as well: a build-up, low level servo control (inner loop, slave-loop) and human operator supervisory control: (outer loop, master-loop). The block diagram of the fuzzy-cascade control loop is in Fig. 6.

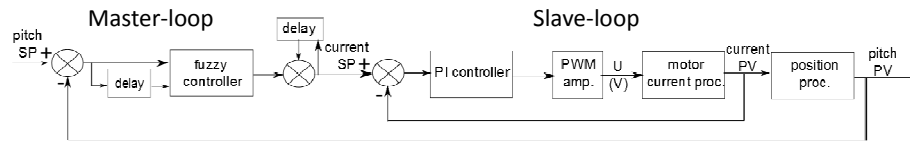


Fig. 6. Block diagram of the cascade vertical position control, fuzzy PI-controller in the outer loop

The master controller is a fuzzy PI-controller where the error input linguistic variable is the difference between the reference position (set-point) and the actual position (pitch). The output linguistic variable, MV is the required change in motor current. The output of the fuzzy controller is added to the previous set-point of the PI-controller (see Fig. 5). The rule-base is symmetric for a diagonal, and has the same structure as it can be seen in Table I. In all presented experiments the applied fuzzy inference method was a max-prod inference and the defuzzification method was the Center of Area (CoA). The fuzzy control mechanism is detailed in literature, for example in [9].

The requirements of control have been set as: avoiding unstable states: growing of position in form of sinusoids is not allowed. Max 20% overshoot and the settling time must be reached in the half-period time of the set-point frequency.

The original control arrangement presented in Fig. 6 has constant proportional gain in the inner loop. Unfortunately this constant parameterization was only good enough for disturbance rejections. For reference tracking the control was either too slow (with low proportional gain, see Fig. 7) or was unstable in positions below the horizontal positions like in Fig. 8. The corresponding phase-plane trajectories can be seen in Fig. 9.

3.3 Development an adaptive fuzzy cascade control loop

In order to achieve better control performance, for appropriate settling time, the proportional gain of the inner loop controller must be low when the process variable is around the set-point to avoid swinging, but must be high when the process variable is located far from the set-point. To solve the problem, an adaptive component has been added to the inner loop. This adaptation based on experiments and has been formulated in form of a fuzzy Decision System (DS).

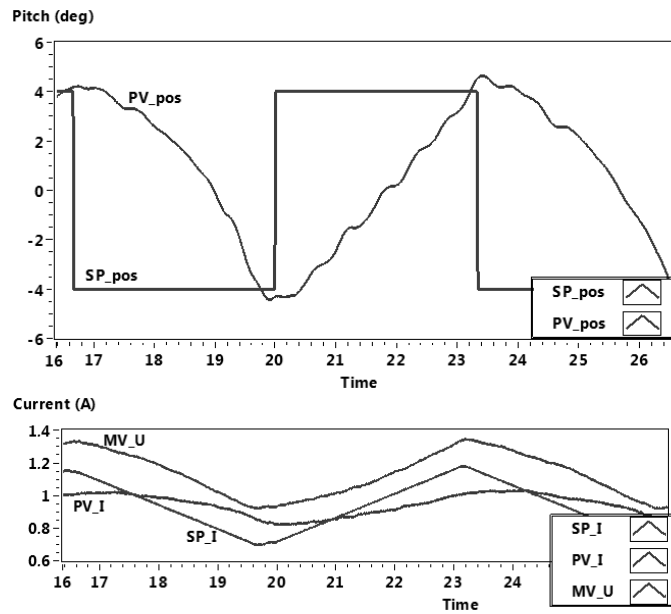


Fig. 7. Set point tracking in the master loop (top) and in the slave loop (bottom). Too slow controls: the required pitch position could not be stabilized

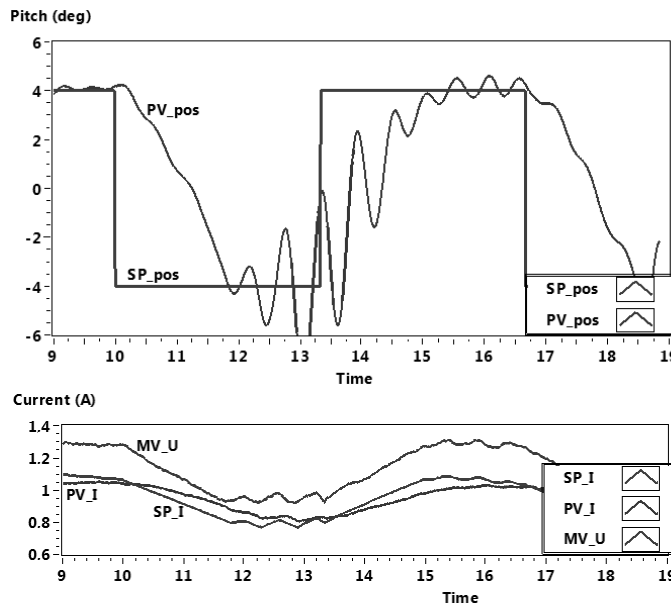


Fig. 8. Set point tracking in the master loop (top) and in the slave loop (bottom). The slave-loop reference tracking is speeded up, but unstable lower position in the master-loop

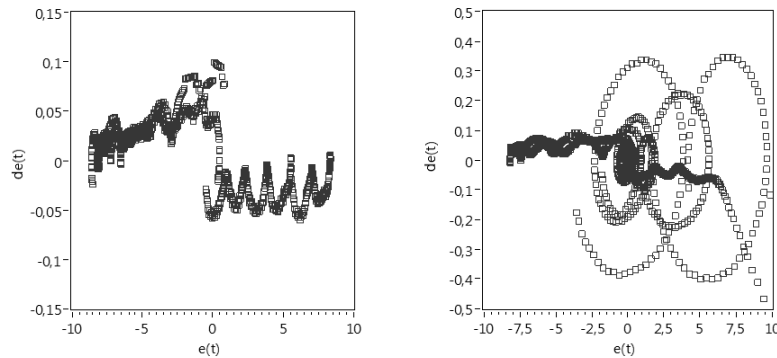


Fig. 9. Phase-plane trajectories of master loops. Left: too slow control: fix point could not be reached in desired time (Time diagram in Fig. 7). Right: Circles mean swing in time domain. Limit cycles represent the unstable characteristic (Time diagram in Fig. 8)

This fuzzy system has two inputs: the error and the windowed error change. Windowed error change means that the discrete derivative of the error has been calculated over a wider time interval in order to better expressing the trend of the change direction and amplitude. The output of the system is the proportional gain of the PI-controller. The rule-base represents the experience gathered in manual tuning procedures and consists of 25 rules. For example: IF Err = Neg AND D_err = Neg THEN Kp = High. IF Err = H_Poz AND D_err = H_Poz THEN Kp = Very_High. IF Err = Zero THEN Kp = Low. Fig. 10 shows the input/output surface of the fuzzy DS. The block diagram of the adaptive control loop can be seen in Fig. 11.

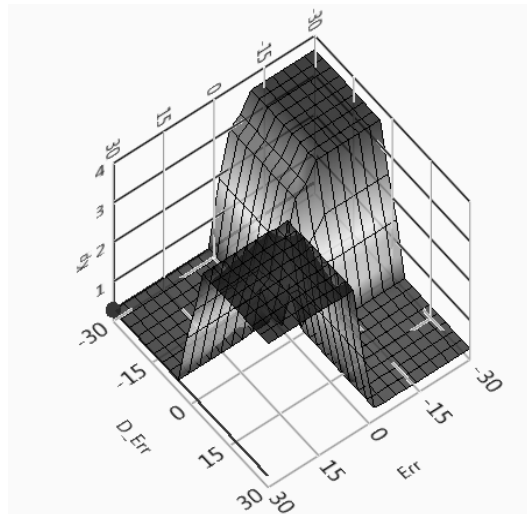


Fig. 10. Input/output surface of the fuzzy DS

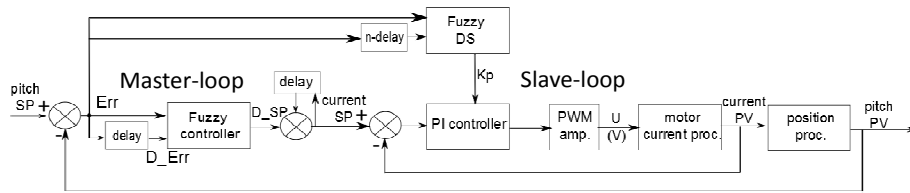


Fig. 11. Block diagram of the cascade vertical position control, adaptive fuzzy PI-controller in the outer loop

The control system has been improved with the presented adaptive parameterization: the control became stable in both positions: below and above the horizontal, see Fig. 12. The control is speeded up and the process variable (pitch) remains in the dynamic range.

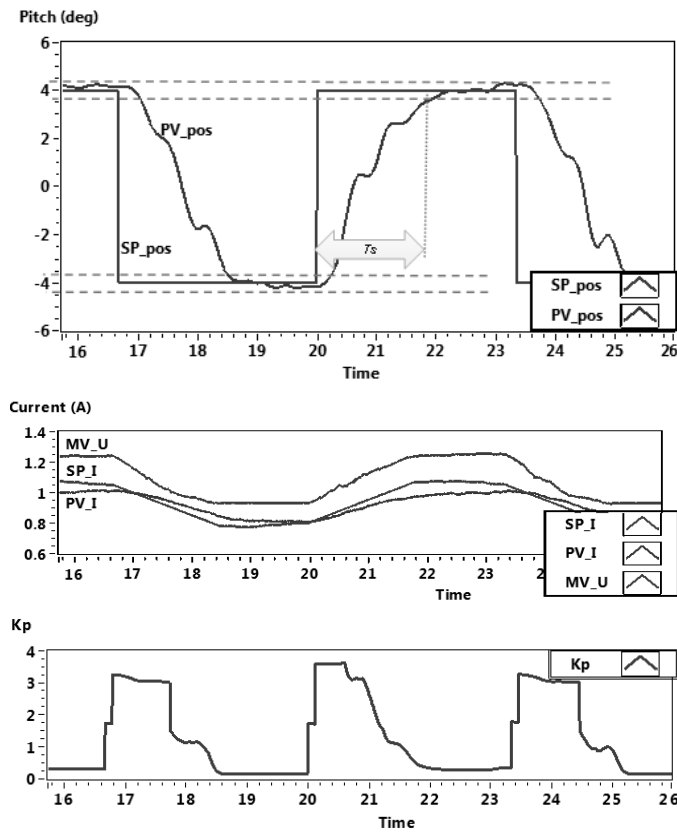


Fig. 12. Set point tracking in the master loop (top), in the slave loop (middle) and evolution of the proportional gain in the inner loop (bottom)

In further, the time-diagram based improving of the control properties is difficult, since the control seems to be satisfying the control criteria. Analyzing the phase-plane

in Fig. 13 suggests that the reference tracking suffers some unwanted swinging. The trajectories approximate the origin on roundabout way. The swing is non-symmetric: on the left hand side of the origin, above negative errors there are more waves (negative error means that the position is over the reference). This fact implies corrections: breaking the symmetry in the fuzzy-controller. Two types of corrections have been tested. First the input partitions were modified in order to be different over the negative and the positive error domains. Then, secondly, the rule base has been modified with the concept that the controller strive to raise the head of the arm, instead of do nothing in negative error situations. The output variable has seven terms in order to refine the controller action. In Table II the modified rule block can be seen.

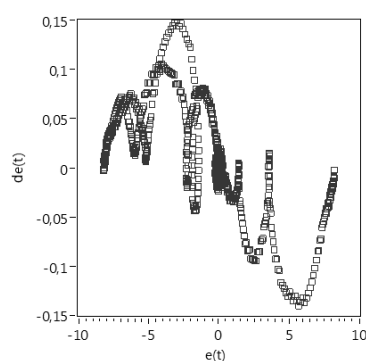


Fig. 13. Phase plane of reference tracking in Fig. 12

Table II

Rule base of a modified fuzzy PI-controller

Δe	NB	N	ZE	P	PB
NB	NEGB_B	NEGB	NEGB	ZERO	POZ
N	NEGB	NEG	NEG	ZERO	POZ
ZE	NEG	ZERO	ZERO	POZ	POZB
P	NEG	ZERO	POZ	POZB	POZB_B
PB	POZ	POZB	POZB_B	POZB_B	POZB_B

Fig. 14 shows the time diagram of the first modification and Fig. 15 shows the time diagram of the second modification. In Fig. 16 the phase planes can be seen.

The settling times are highlighted in the time diagrams. It can be seen, that the modification of the membership functions increased the settling time, i.e. the control became slower. On the other hand, in the second case, the control is speeded up particularly in reference tracking at positive position.

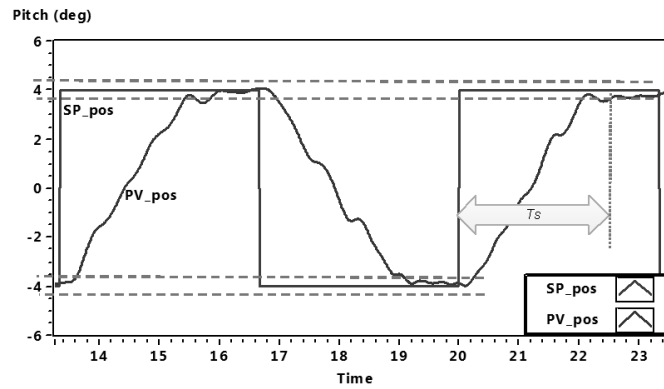


Fig. 14. Set point tracking in the master loop. The settling time is increased

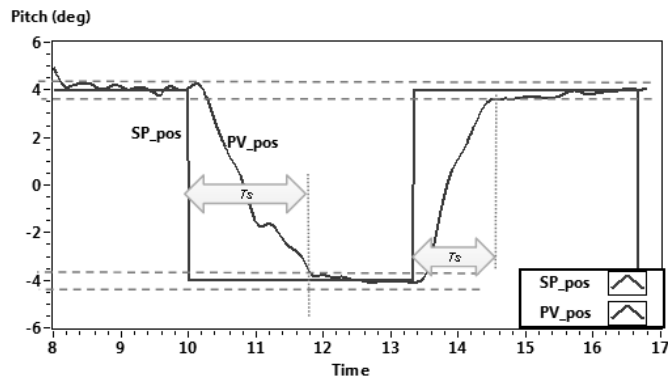


Fig. 15. Set point tracking in the master loop. Different settling times at positive and negative pitch positions. The positive side speeded up, the negative side stabilized

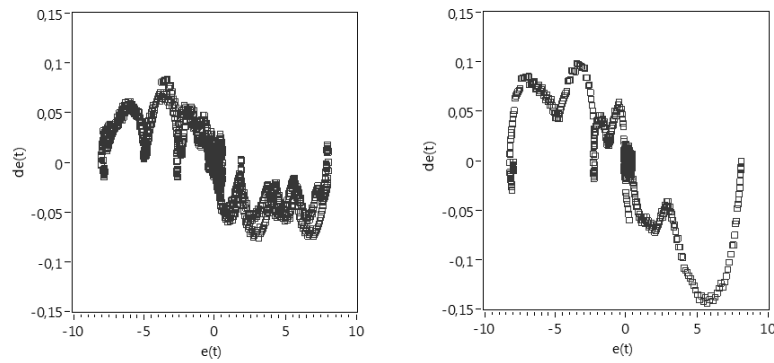


Fig. 16. Phase planes of reference trackings in Fig. 14 (Left) and Fig. 15 (Right)

Compared the phase-planes, the first case shows more waves and movements in trajectories than the second case. The phase-plane of the second case approximates best to the disturbance-free properly parameterized control model trajectories in *Fig. 4*. The upward and downward movements have different transients, since mowing down requires smoother intervention in order to avoid unstable situations. In *Table III* the time data are compared. With the proposed settings both the settling time and the rising time could be reduced without overshoot. This parameterization is acceptable and the design procedure ends. It can be concluded, that the phase-plane investigation helps to develop a correct control structure, and the phase-plane trajectories support the validation of a fuzzy controller.

Table III

Settling times and Rise times in the master loop in unit time (ut)

Fuzzy cascade control type	k_p	Settling time (ut)	Rise time (90%) (ut)	Settling time (ut)	Rise time (90%) (ut)
		positive position		negative position	
symmetric rule base	constant, low value	no in desired time interval			
symmetric rule base	constant, high value	no in desired time interval		no (unstable)	
symmetric rule base	adaptive	1.9	1.7	1.9	1.7
asymmetric membership function distribution	adaptive	2.6	2.0	2.6	2.0
asymmetric rule base	adaptive	1.2	1.0	1.8	1.6

4. Conclusion

In this paper a novel, phase plane trajectories based fuzzy controller design procedure has been presented. Not only a fuzzy controller but a fuzzy-DS has been implemented in a cascade control loop. The fuzzy DS determines the proportional gain of the classic PI-controller in the inner loop. Experience based construction of an adaptive gain could be much faster than defining a precise analytical function. Developing a fuzzy controller is always an iterative process, and with the help of investigation of phase-plane trajectories, this process could be shortened. The phase-plane trajectories could suggest targeted modifications in the fuzzy controller and furthermore the phase-plane trajectories support the validation of a designed fuzzy controller as well.

Acknowledgements

This work has been undertaken as a part of a project TÁMOP-4.2.2.C-11/1/KONV-2012-0005. III./6.

The present scientific contribution is dedicated to the 650th anniversary of the foundation of the University of Pécs, Hungary.

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